IE 327 - Facility Design & Material Handling
Final Project Report

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a) Introduction & Project Scope

The Purpose of the project is to analyze the current workflow inside the Madina Pharmacy in order to improve its overall efficiency. In order to do this, our group collected data & observations, analyzed the workflow, and identified improvements for the pharmacy's layout. Currently, the pharmacy has an inefficient layout according to facility design principles from our class. The project aims to improve these inefficiencies using the various techniques and methods taught in class.

The Project scope begins with analyzing the current layout of the pharmacy and workflow, redesigning it using our analysis from visiting it. We improve this by turning the current workflow into a structured assembly line. By making the flow linear, we avoid the workflow moving back and forth that creates inefficiencies. One of the identified inefficiencies was congestion in key areas (Such as the counter, etc). Finally, we will use flow and distance data to support the redesign.

b) Summary of Data Collected

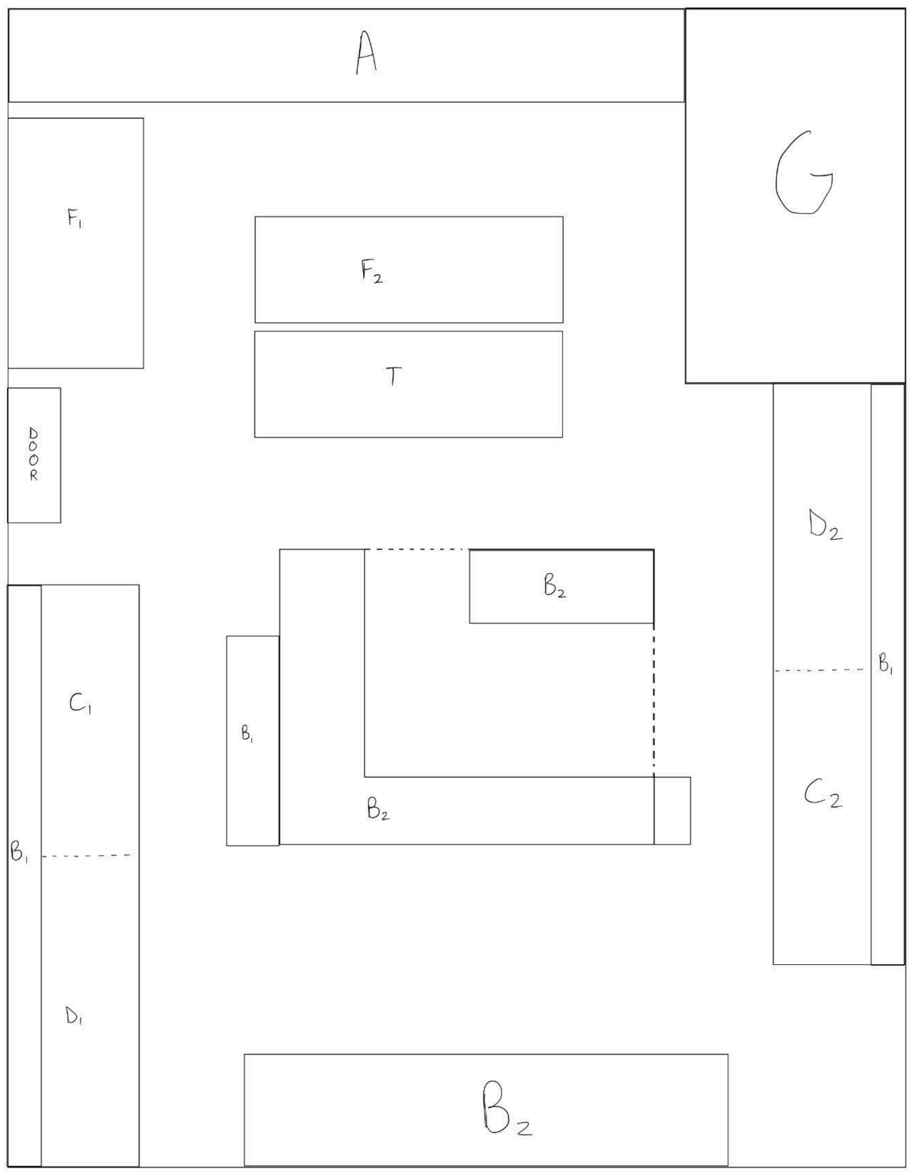
Our group made progress in collecting some of the foundational data for the pharmacy final design project. We obtained detailed measurements of the pharmacy, including the dimensions of storage areas, desks, shelves, and racks. These measurements that were gained have allowed us to start constructing an accurate layout of the current system of the pharmacy. Some notable measurements and data:
Height of 3.32 meters for the store area, 3.33 meters for the pharmacy.
The whole pharmacy is currently organized into distinct zones, such as the Store/shopping and waiting area, Prescription processing area, product storage, and the packaging/delivery sections. Our group also collected data on the equipment inventory, finding a total of 8 shelves, 4 racks, and 1 push cart available.

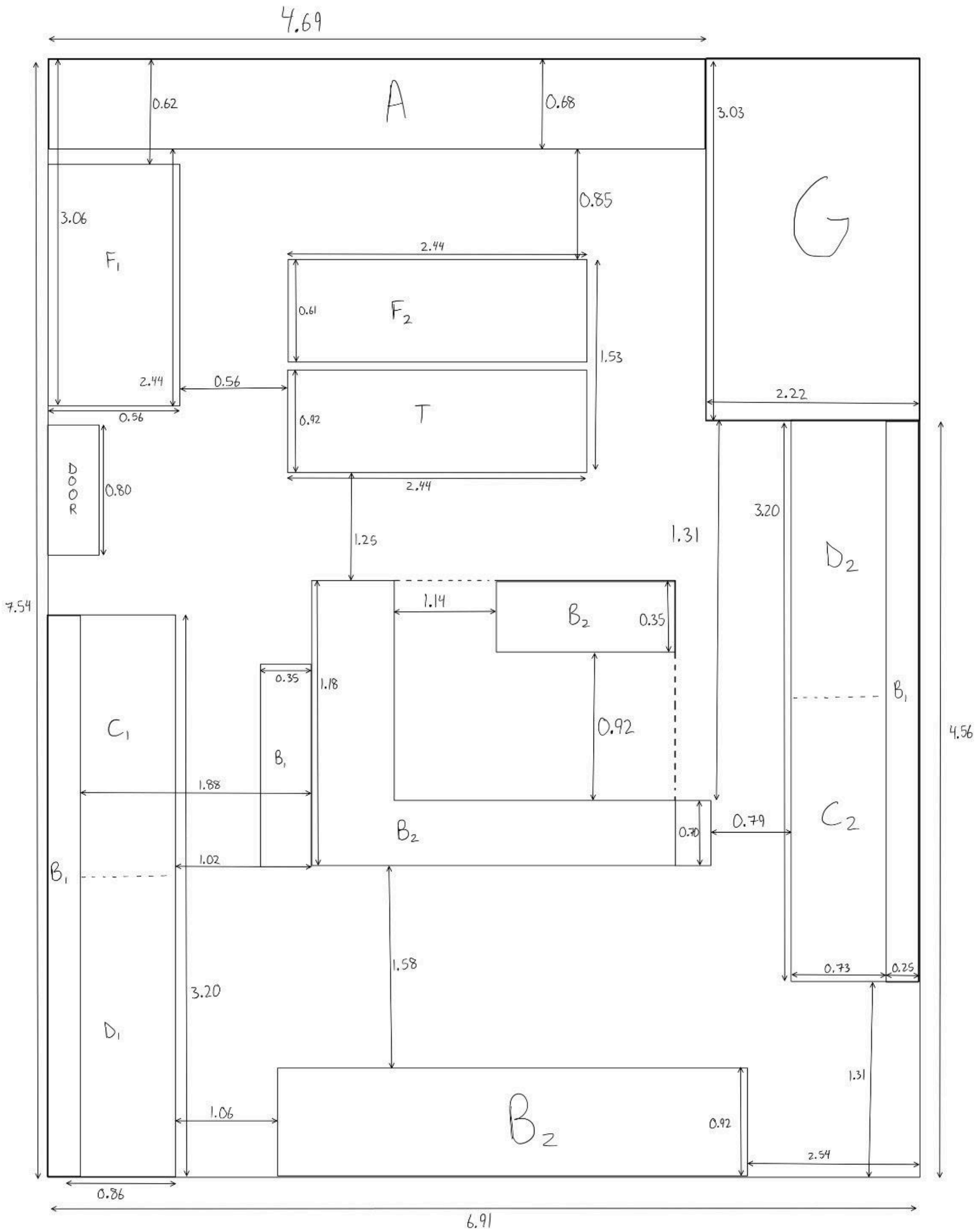
c) Current Facility Layout Analysis

The current facility layout shows that the pharmacy is organized into separate work areas, but the stations are not arranged in the best order for prescription flow. The current system causes employees to move between storage, filling, verification, and packaging in a non-linear path. This creates backtracking and increases the amount of walking needed to complete prescriptions.

The current layout also has congestion near packaging and pickup because completed prescriptions collect in those areas before being picked up by customers or delivery drivers. Employee movement and activities facing customers are close to each other, which creates crossing paths and makes the space feel crowded (bottlenecks).

Even though the current layout works, it does not support an assembly line process. The main issue is that high-flow stations are not close enough together, which increases the flow-distance cost.

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DOOR
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D2
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C2
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B1
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D1
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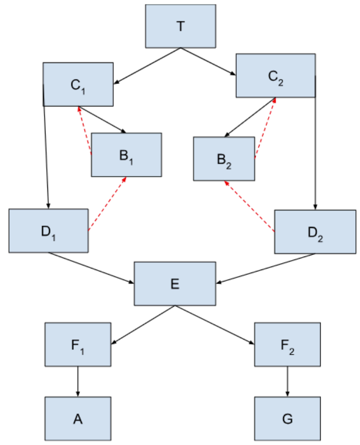
d) Pharmacy Flow Analysis



Symbol/Station	Description
T- Technician (label printing and data processing)	Starts the process by entering prescription information and printing labels.
B_{1A} /B_{1B}- Fast-moving medication	Stores medications used most often. These are kept close to the filling stations.
B_2- Slow-moving medication	Stores medications used less often. These can be further from the main filling area.
C_1/C_2 - Filling Stations	Fillers retrieve medications, count pills, and prepare prescriptions for verification.
D_1/D_2- Pharmacist Verification Stations	Pharmacists check the filled prescription for accuracy before it moves forward.
E- Packing/Bagging	Verified prescriptions are bagged and prepared for pickup or delivery.
F_1- Pickup Storage	Holds completed prescriptions that customers will pick up at the counter
F_2- Delivery Staging	Holds completed prescriptions that are waiting for delivery drivers.
A- Order Intake	Customers receive their prescriptions here after the staff retrieves them from pickup storage.
G- Delivery Driver Pickup	Delivery drivers collect prescriptions from staging before leaving for deliveries.

The system was estimated with calculations at 24 prescriptions per hour. Each filler processes about 12 prescriptions per hour at the two stations. Based on the 60% fast-moving and 40% slow-moving split, each filler handles about 7 fast-moving prescriptions per hour and 5 slow-moving prescriptions per hour.

After filling, the prescriptions move to a pharmacist for verification. Both pharmacists then send verified prescriptions to the packaging. From packaging, prescriptions are split into pickup and delivery. Since 65% is pickup and 35% is delivery, about 16 prescriptions per hour go to pickup and 8 prescriptions per hour go to delivery.

e) Data Collection Methods

Our group used direct observation, time estimates, and layout measurements to collect the data needed for the analysis. During the visit, we observed how prescriptions moved between stations and how employees interacted with the layout. We also recorded the general process steps and estimated how often prescriptions moved between departments. The methods used helped us calculate the flow- distance cost of the current layout and the redesigned layout cost. The main data collection methods were:

- Observing prescription movement between stations
- Estimating prescriptions processed per hour
- Separating fast-moving and slow-moving medication flows
- Measuring distances between departments using the layout
- Using centroid points to calculate travel distances
- Building a flow matrix and a distance matrix

f) Current Flow Matrix

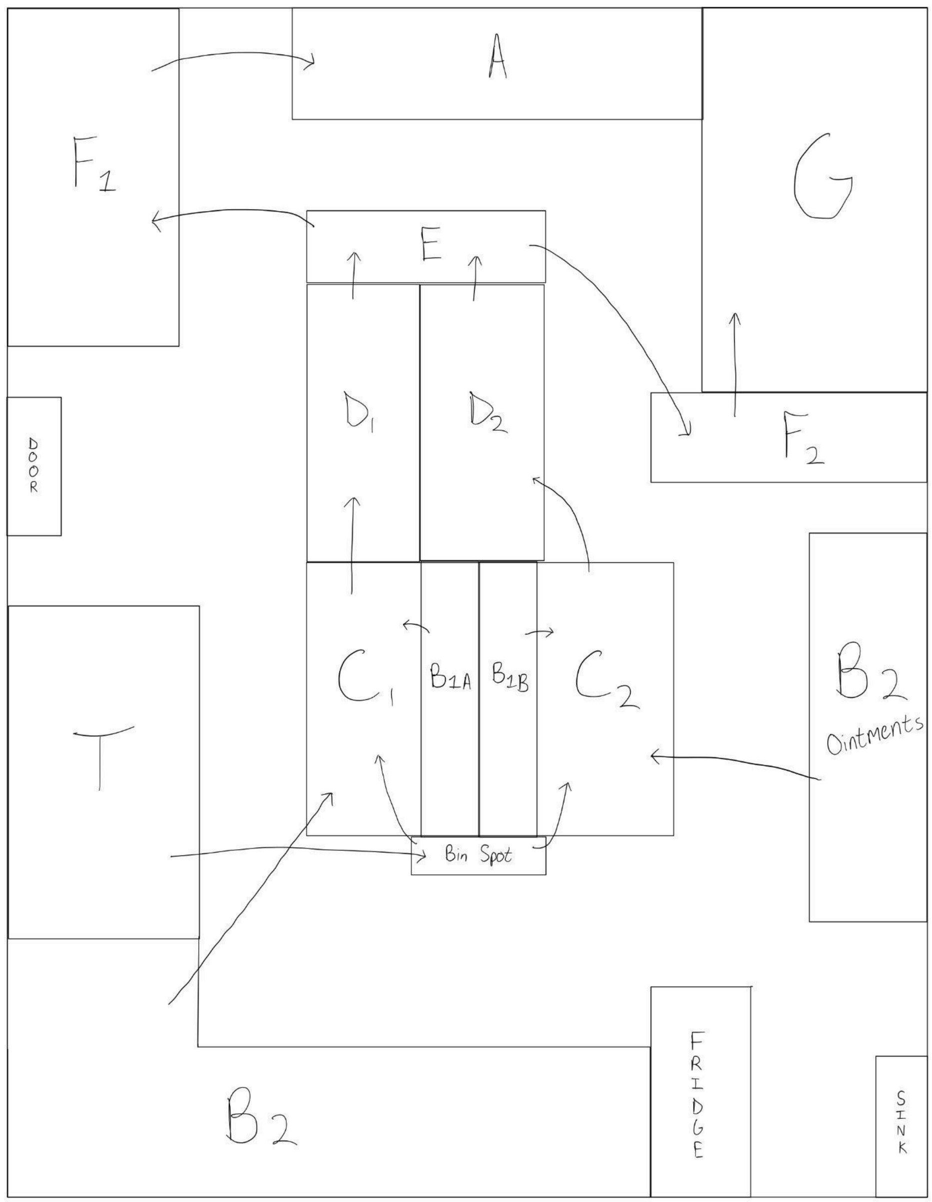
From/ To	T	C_1	C_2	B_{1A}	B_{1B}	B_2	D_1	D_2	E	F_1	F_2	A	G
T	-	12	12	0	0	0	0	0	0	0	0	0	0
C_1	0	-	0	7	0	5	12	0	0	0	0	0	0
C_2	0	0	-	0	7	5	0	12	0	0	0	0	0
B_{1A}	0	7	0	-	0	0	0	0	0	0	0	0	0
B_{1B}	0	0	7	0	-	0	0	0	0	0	0	0	0
B_2	0	5	5	0	0	-	0	0	0	0	0	0	0
D_1	0	0	0	1	0	1	-	0	12	0	0	0	0
D_2	0	0	0	0	1	1	0	-	12	0	0	0	0
E	0	0	0	0	0	0	0	0	-	16	8	0	0
F_1	0	0	0	0	0	0	0	0	0	-	0	16	0
F_2	0	0	0	0	0	0	0	0	0	0	-	0	8
A	0	0	0	0	0	0	0	0	0	0	0	-	0
G	0	0	0	0	0	0	0	0	0	0	0	0	-

From Flow-To-Flow Matrix:

- 24 prescriptions/hours total
- Each filler approx 12 prescriptions/hour
- 60% fast movers & 40% slow movers
- 65% pickup & 35% delivery

Key Flows:

- T → C1/C2 = 12
- C → D = 12
- D → E = 24
- E → Pickup = 16
- E → Delivery = 8


A
F1
G
E
D1
D2
F2
DOOR
C1
B1A
B1B
C2
B2
Ointments
T
Bin Spot
FRIDGE
SINK
B2

g) Current Flow -Distance Analysis

From/ To	T	C_1	C_2	B_{1A}	B_{1B}	B_2	D_1	D_2	E	F_1	F_2	A	G
T	-	4.45	7.41	3.43	7.52	3.47	6.05	5.81	3.94	2.82	0.77	2.27	4.55
C_1	4.45	-	6.78	1.02	6.89	2.84	1.60	7.06	3.97	3.45	5.22	6.72	9.00
C_2	7.41	6.78	-	6.46	0.11	3.94	7.06	1.60	4.95	10.23	8.18	9.66	5.04
B_{1A}	3.43	1.2	6.46	-	6.57	2.52	2.62	6.04	2.65	3.77	4.20	5.70	7.98
B_{1B}	7.52	6.89	0.11	6.57	-	4.05	7.17	1.71	5.06	10.34	8.29	9.77	5.15
B_2	3.47	2.84	3.94	2.52	4.05	-	4.22	4.44	2.11	6.29	4.24	5.72	6.38
D_1	6.05	1.60	7.06	2.62	7.17	4.22	-	8.66	2.11	5.05	6.82	8.32	10.60
D_2	5.81	7.06	1.60	6.04	1.71	4.44	8.66	-	6.55	8.63	6.58	8.06	3.44
E	3.94	2.97	4.95	2.65	5.06	2.11	2.11	6.55	-	6.42	4.71	6.21	8.49
F_1	2.82	3.45	10.23	3.77	10.34	6.29	5.05	8.63	6.42	-	2.07	3.57	5.85
F_2	0.77	5.22	8.18	4.20	8.29	4.24	6.82	6.58	4.71	2.07	-	1.50	3.78
A	2.27	6.72	9.66	5.70	9.77	5.72	8.32	8.06	6.21	3.57	1.50	-	4.63
G	4.55	9.00	5.04	7.98	5.15	6.38	10.60	3.44	8.49	5.85	3.78	4.63	-

Using the equation: $\text{Total Cost} = \sum(\text{Flow}_{i\square} \times \text{Distance}_{i\square})$

The flow values came from the current flow matrix, and the distances were based on the current layout measurements and centroid locations of each department. This identifies how much travel is created by the current layout and is used to compare it to the redesigned layout.

For the current layout, the total Flow × Distance cost = 608.97

The current layout creates a large amount of extra movement between stations. The highest movement areas are between storage, filling, verification, packaging, pickup, and delivery. Since these areas have frequent prescription movement, the distance between them has a large impact on the total cost.

The current layout creates extra travel because employees have to move back and forth between storage and filling areas. The redesigned layout will need to have high-flow stations close together and create a better process.

h) **Pharmacy Workflow Bottleneck Analysis**

The main bottleneck is pharmacist verification because every prescription must go through verification before it can be packaged and moved to pickup/delivery. If verification is slow, it can limit the entire process.

Another bottleneck is the backtracking between storage and filling. Fillers often move between the filling station and medication storage. If storage is not close to the filling stations, this creates extra walking and slows the process. Packaging and pickup are also congested because completed prescriptions build up. There are layout issues, such as phone calls interrupting workers while they are filling out prescriptions. This slows order intakes and creates a bottleneck on the front-end side.

i) **Redesigned Layout Proposal**

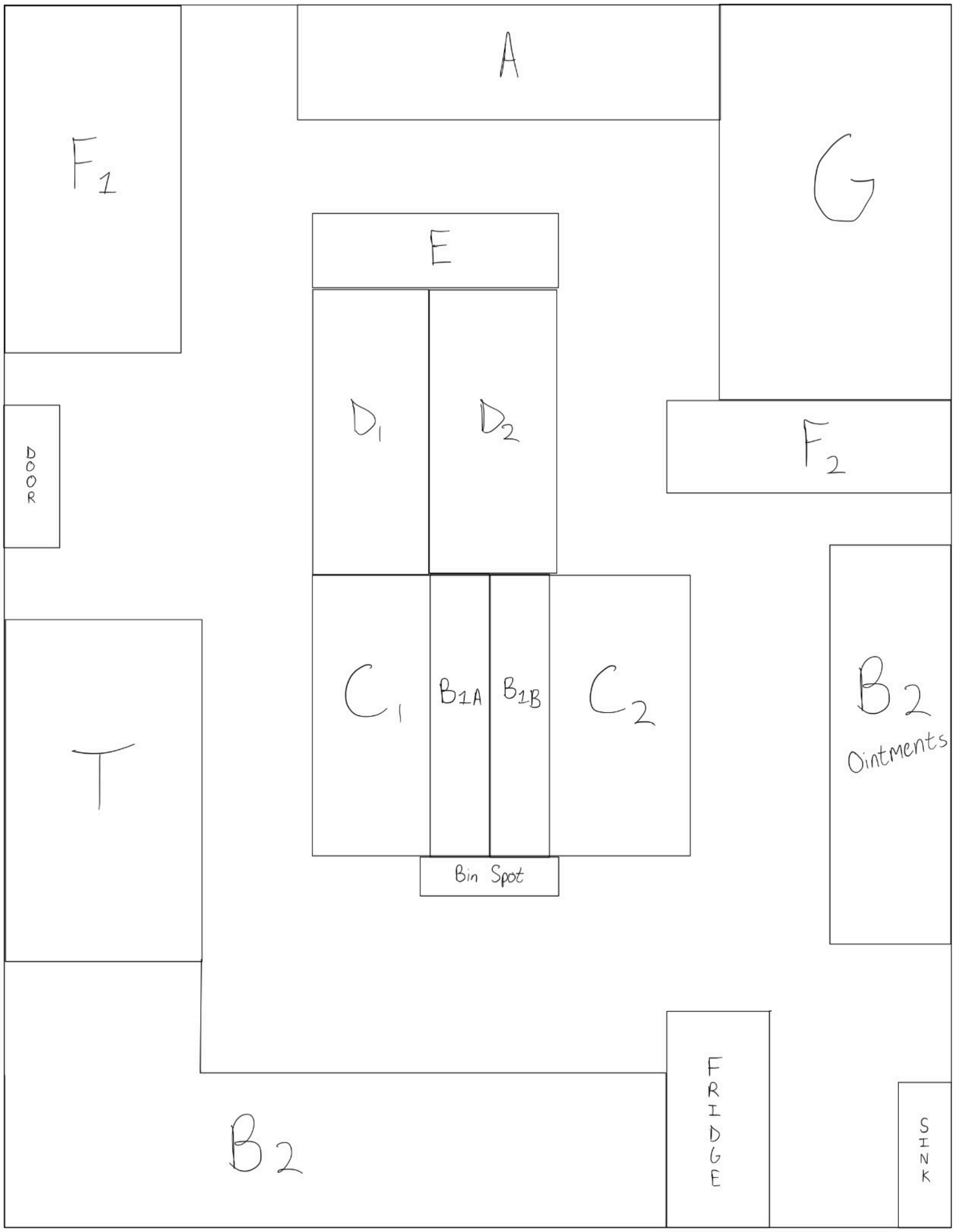
The redesigned layout was developed to transform the current workflow into a structured assembly line. This would help reduce backtracking and improve the overall efficiency of the facility. The main objective was to create a linear flow from order intake to final pickup or delivery.

In the proposed layout, the stations are reorganized so that the prescription processing follows a clear forward progression. The process begins at the technician station (T), where orders are entered and labels are printed. From there, prescriptions move directly to the filling stations (C1 and C2), which are now positioned centrally and adjacent to fast-moving medications storage (B1A and B1B). This reduces the distance traveled when retrieving commonly used medications.

The pharmacist verification stations (D1 and D2) are placed directly above filling stations, ensuring a smooth vertical transition between stages. After the verification process is complete, prescriptions move upward to the packaging area (E), which is centrally located to distribute complete prescriptions efficiently.

From the packaging station, prescriptions are separated into pickup (F1) and delivery staging (F2). These areas are positioned closer to their respective endpoints (Customer counter A and delivery driver pickup G), minimizing unnecessary movement. Slow-moving medication storage (B2) is placed along the outer edges of the layout due to its less frequent traffic attraction and does not need to be in close proximity to the main workflow.

Overall, the redesigned layout eliminates unnecessary crossing and organizes all major stations into a continuous, forward-moving, consistent with an assembly line approach.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G
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DOOR
F2
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B1A
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Ointments
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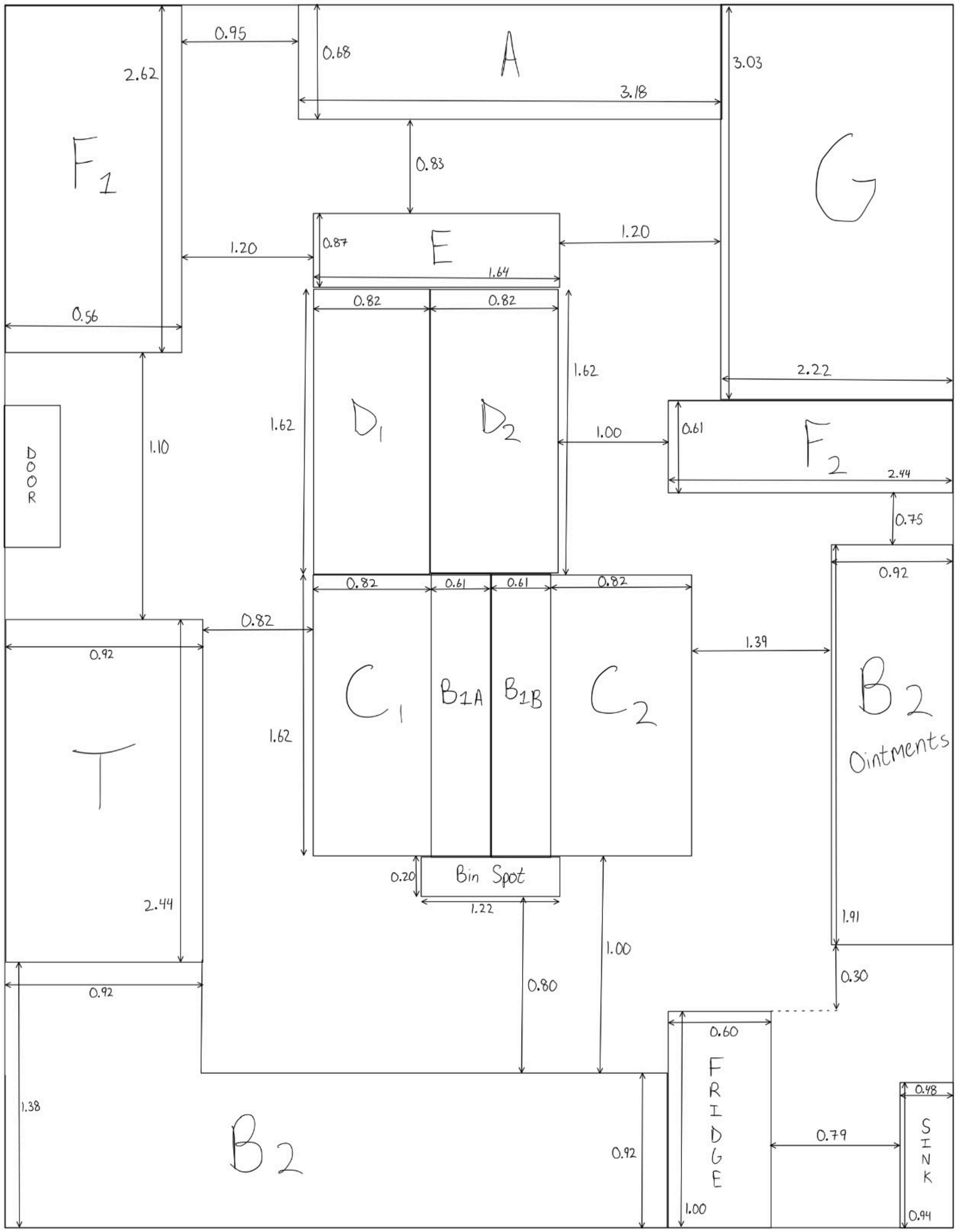
Bin Spot
FRIDGE
SINK
B2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F1
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G
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E
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1.20
0.82
0.82
D1
D2
1.62
1.62
1.00
F2
0.61
2.44
0.75
DOOR
1.10
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0.48
SINK
0.94

j) Redesigned Flow Distance Analysis

The redesigned layout was evaluated using a Flow x Distance analysis to quantify the efficiency improvements. This analysis was conducted using centroid-based distance calculations, where a central point represented each department. Distances between departments were measured using the updated layout dimensions.

The flow values between departments were obtained from the previously developed flow matrix, which reflects the movement of prescriptions between stations. These flow values were multiplied by the corresponding centroid distances for each pair of departments.

The total flow x distance cost for the updated layout was calculated using the following relationship:

$$\text{Total Cost} = \sum(\text{Flowij} * \text{Distance ij})$$

Based on the centroid calculations and updated layout, the total flow x distance value for the redesigned layout system was:

Total Flow x Distance = 392.12

This value represents the total weighted travel distance of prescriptions throughout the system. The reduced distances between high-flow stations (Filling, Verification, and Packaging) significantly contribute to lowering this value.

k) Layout Principles

The new design layout aligns well with key facility design principles that will help improve the facility's operational efficiency.

Firstly, the optimized layout minimizes the travel distance by placing high-flow departments closer to each other. Stations such as filling (C1, C2), verification (D1, D2), and packaging (E) are positioned adjacently, reducing all unnecessary movement between the stages.

Secondly, the design improves the flow efficiency by creating a linear and forward-moving process. Unlike the original layout that required backtracking between stations, the new layout ensures that the prescriptions move in one direction from start to finish. This eliminates any crossing paths and reduces any congestion.

Next, the layout reduces bottlenecks by organizing work stations based on process sequence. The packaging area is centrally located, which allows it to efficiently distribute prescriptions for pickup and delivery without causing buildup in one location. The layout also improves space utilization by placing slow-moving storage (B2) along the outer edges, freeing up central space for high-frequency operations. This will now ensure that frequently used areas remain accessible and efficient.

Finally, the separation of customer-facing areas and internal workflow reduces interference between employees and customers, improving both safety and productivity.

l) Results and Improvements

The new layout results in an efficient and organized workflow compared to the current system. By converting the current layout into an assembly line structure, the overall movement of prescriptions becomes more streamlined, reducing unnecessary travel and backtracking. The Flow x Distance analysis of the redesigned layout resulted in a total cost of **392.12**.

Percent Improvement = [(608.97 − 392.12) / 608.97] × 100 Percent Improvement = 35.6%

The redesigned layout reduced the total movement cost by about 35.6%. The improvement comes mainly from placing filling, verification, and packaging closer together and reducing backtracking between storage and filling.

From an operational standpoint, the new layout reduces congestion in key areas such as packaging and pickup by evenly distributing workflow. With fewer interruptions, employees are able to complete tasks more quickly, which can lead to faster processing times and an increased throughput. In addition, the clearer better use of station location improves coordination between technicians and pharmacists, helping reduce delays at bottleneck points such as verification. Overall, the redesigned system provides a more efficient, organized, and scalable solution that better supports the pharmacy's operational needs.

m) Group Contributions

Paul Poleon – Led the team reports, organized meetings, and performed distance calculations
Alexander Do – Measured layout and redesigned floor plan with updated measurements
Andrew Pokornicki – Developed product flow paths and defined stations
Carlos Aceves – Built flow matrix and aided in calculations
Jason Jackson – Analyzed employee movement and congestion

T	0.46	2.66
C1	2.15	2.73
C2	4.19	2.73
B1A	2.865	2.73
B1B	3.475	2.73
B2	3.30	1.12
D1	2.15	4.35
D2	2.97	4.35
E	2.58	5.595
F1	0.28	6.23
F2	5.69	4.205
A	2.35	7.2
G	5.8	6.025

Second design

T	2.34	4.94
C1	0.43	2.4
C2	6.55	1.74
B1A	1.1	2.75
B1B	6.66	1.74
B2	3.16	2.29
D1	0.43	0.8
D2	6.55	3.34
E	2.17	1.17
F1	0.28	5.7
F2	2.34	5.71
A	2.35	7.2
G	5.8	6.025

First design